

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
April - May 2015

Max. Marks: 100

Class: SE

Name of the Course: Automata Theory

Course Code: UITC404

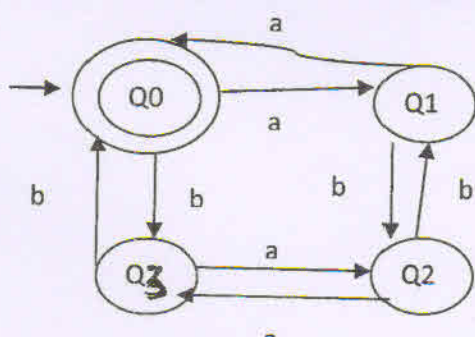
Duration: 03 Hrs

Semester: IV

Branch: IT

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q1 (a)	Write the steps of Mathematical Induction. Use the steps to prove the following: $1+3+5+\dots+2n-1=n^2$	05
Q1 (b)	Define Pigeon Hole Principle and Pumping Lemma. Prove their equivalence.	05
Q2 (a)	Design Mealy machine for incrementing the value of any binary number by one. The output should also be a binary number with value one more than the number given. OR Differentiate between Moore and Mealy machine. Design Moore machine to convert each substring abb to aba over the $\Sigma(a,b)^*$.	10
Q2 (b)	Design a DFA to accept (a) Binary string to accept all strings in which every 1 is followed by 00 (b) Strings over $\Sigma(a,b)^*$ with an odd number of a's and odd number of b's	10
Q3 (a)	Describe the language accepted by the following regular expression and draw corresponding NFA. (1) $(a+b)^*aba$ (2) $(0+1)^*(00+11)$ (3) $1^*01^*01^*$ (4) $10+(0+11)0^*1$ (5) $(aa)^*b(bb)^*$	10
Q3 (b)	Find Regular Expression for the following Finite automata  OR Prove using Pumping Lemma, $L = \{a^i b^{2i} \mid i \geq 1\}$ is not regular.	10

Q4 (a)	Explain steps to convert CFG into equivalent CNF with an example. Convert the given CFG productions into Greibach normal form $S \rightarrow ABA AB BA AA A B$ $A \rightarrow aA \mid a$ $Q \rightarrow bB \mid b$	10
Q4 (b)	Consider the given CFG productions $S \rightarrow aSb bSa SS \epsilon$ For string abbaabab find - Leftmost derivation - Rightmost derivation - Comment whether given grammar is regular or not - Using parse tree prove whether the grammar is unambiguous or not	10
Q5 (a)	Construct a Pushdown Automata (PDA) for the following language $L = \{ a^{3n}b^n \mid n > 0 \}$ Make sure the PDA is deterministic OR Convert the given CFG to an equivalent PDA $S \rightarrow aAA a$ $A \rightarrow bS \mid aS$	10
Q5 (b)	Design Turing machine for language of even palindrome over $\Sigma (0,1)^*$ OR Define Turing Machine. Explain Multitape and Multihead Turing Machine.	10
Q6	Write short note on - Recursive Languages and Recursively Enumerable Languages - Computability and Decidability OR State and explain POST Correspondence Problem.	10